

SAN LUIS VALLEY, CO, USA

WMO: 724620

Lat: 37.439N

Lon: 105.861W

Elev: 2296

StdP: 76.62

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23061

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.1	-28.2	0.4	-24.2	-25.1	0.5	-19.5	12.3	2.8	11.0	2.0	1.3	260	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	30.0	12.8	28.7	12.5	27.4	12.4	15.2	24.5	14.7	23.7	14.1	22.8	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.8	12.2	16.2	12.0	11.6	16.1	11.2	11.0	16.2	50.4	24.4	48.7	23.7	47.0	22.6	18.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.0	DB	-29.2	32.3	4.6	1.0	-32.4	33.1	-35.1	33.7	-37.7	34.3	-41.0	35.0	(4)
(5)			WB	-29.4	16.6	4.4	0.8	-32.6	17.1	-35.2	17.5	-37.6	18.0	-40.9	18.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-7.8	-4.0	1.8	5.8	10.9	15.9	18.5	17.2	13.3	6.6	-0.6	-7.1
	DBStd	9.88	5.48	5.24	4.18	3.64	3.52	2.49	1.77	1.72	2.92	3.82	4.71	5.32
	HDD10.0	2332	552	393	253	129	33	1	0	0	8	116	318	529
	HDD18.3	4564	811	626	511	375	231	77	19	42	152	364	568	788
	CDD10.0	848	0	0	0	4	60	179	264	224	106	11	0	0
	CDD18.3	39	0	0	0	0	0	6	25	8	1	0	0	0
	CDH23.3	1838	0	0	0	1	77	531	682	411	134	2	0	0
	CDH26.7	427	0	0	0	0	9	135	195	76	12	0	0	0
Wind	WSAvg	3.2	2.3	3.0	3.8	4.6	4.3	3.9	2.9	2.7	2.9	3.0	2.6	2.3
Precipitation	PrecAvg	191	6	8	12	12	15	15	28	31	22	18	11	11
	PrecMax	293	19	36	41	44	47	66	89	83	47	60	31	39
	PrecMin	111	0	0	1	0	0	0	1	5	1	0	0	0
	PrecStd	46	5	7	10	10	12	13	19	19	13	16	9	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	13.8	19.8	22.4	27.9	31.7	31.8	30.1	28.2	23.5	17.3	11.4
		MCWB	2.4	3.7	6.0	7.4	10.8	12.4	13.8	12.9	12.3	9.2	5.5	2.8
	2%	DB	7.8	11.1	17.1	20.2	25.4	29.7	30.3	28.7	26.4	21.5	15.1	8.5
		MCWB	0.8	2.4	4.9	6.4	9.4	11.8	13.4	13.0	11.5	8.3	4.8	1.4
	5%	DB	5.3	8.7	14.9	18.6	23.6	28.3	28.9	27.4	24.9	19.5	12.6	5.9
		MCWB	-0.5	1.3	4.0	5.7	8.6	11.2	13.3	13.0	10.9	7.6	3.6	0.0
	10%	DB	2.6	5.9	12.4	16.8	21.8	26.7	27.4	25.9	23.3	17.3	10.0	3.2
		MCWB	-1.7	0.0	2.9	5.1	8.0	10.8	13.5	13.0	10.7	6.7	2.4	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	4.4	6.5	8.4	11.8	14.6	16.2	15.9	14.3	11.4	6.6	3.6
		MCDB	9.3	12.0	17.9	19.2	24.1	25.1	24.8	24.9	23.1	19.2	14.8	9.6
	2%	WB	1.2	2.9	5.3	7.2	10.5	13.6	15.5	15.2	13.3	9.8	5.3	1.9
		MCDB	6.4	10.0	15.7	18.1	22.1	24.3	24.8	23.7	21.9	18.0	13.8	7.6
	5%	WB	0.0	1.6	4.3	6.3	9.6	12.8	15.0	14.6	12.6	8.5	4.0	0.3
		MCDB	4.4	7.9	13.8	16.9	20.5	23.8	24.4	22.9	21.0	17.2	11.5	5.2
	10%	WB	-1.3	0.3	3.2	5.4	8.8	11.9	14.5	14.0	11.8	7.4	2.7	-1.2
		MCDB	2.7	5.3	11.6	15.7	19.7	23.0	23.6	22.1	20.1	15.9	9.5	3.3
Mean Daily Temperature Range	5% DB	MDBR	18.7	17.6	18.4	18.2	19.0	20.8	18.7	18.2	19.4	19.7	19.3	18.2
		MCDBR	21.2	20.4	22.5	22.2	22.2	23.4	21.1	20.7	22.1	22.9	22.9	20.6
		MCWBR	14.8	12.7	11.9	10.7	9.2	8.8	7.3	7.6	9.2	11.6	13.6	14.1
	5% WB	MCDBR	19.5	19.0	20.9	20.5	19.8	20.1	17.3	16.6	17.6	20.2	21.4	19.3
		MCWBR	13.9	12.0	11.2	10.1	8.2	7.6	6.1	6.3	7.5	10.8	12.9	13.3
Clear-Sky Solar Irradiance	taub	0.221	0.229	0.260	0.278	0.297	0.302	0.348	0.325	0.286	0.256	0.235	0.226	
	taud	2.615	2.607	2.531	2.508	2.507	2.491	2.427	2.464	2.593	2.651	2.627	2.591	
	Ebn at Noon	1007	1032	1016	1004	983	972	925	943	973	985	979	977	
	Edh at Noon	69	79	95	104	106	108	114	107	89	75	68	66	
All-Sky Solar Radiation	RadAvg	3.09	4.14	5.28	6.31	7.07	7.63	6.68	6.14	5.54	4.38	3.29	2.67	
	RadStd	0.17	0.27	0.36	0.35	0.51	0.40	0.34	0.27	0.29	0.26	0.19	0.15	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+53	+10	
(47) Regional (21 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81	+39	

Nomenclature: See separate page